Domain and Range

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Domain from equations

State the domain of each function:

a $f(x) = \frac{1}{x-4}$

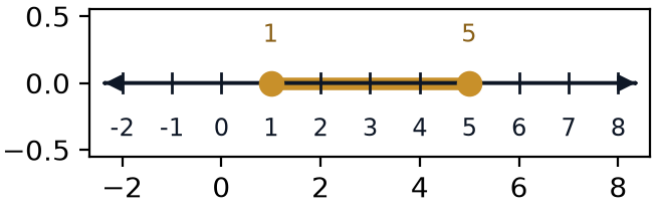
b $g(x) = \sqrt{2x-6}$

c $h(x) = \frac{x+1}{x^2-9}$

d $k(x) = \sqrt[3]{x}$

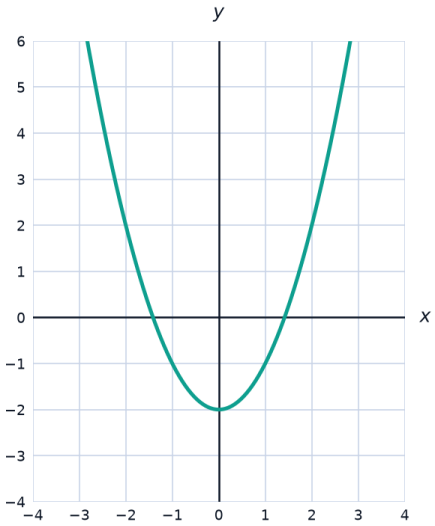
2 [Foundational] • Target: Domain from equations

Find the domain of $f(x) = \frac{\sqrt{5-x}}{x-1}$. Write your answer in interval notation.



3 [Foundational] • Target: Domain and range from graphs

The graph of $y = x^2 - 2$ is shown. State the domain and range of the function.



4 [Foundational] • Target: Finding the range algebraically

By completing the square, find the range of $f(x) = x^2 + 4x + 1$.

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5 [Foundational] • Target: Finding the range algebraically

Consider $f(x) = -2(x-1)^2 + 5$.

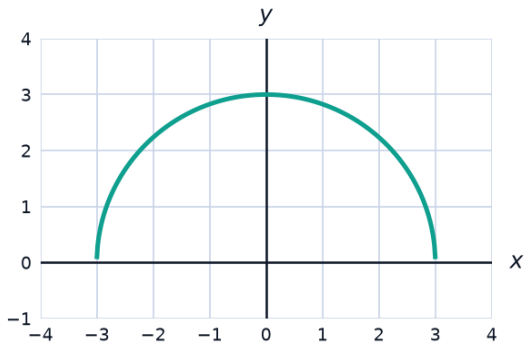
a State the coordinates of the vertex.

b Does the parabola open upward or downward?

c State the range of f.

6 [Foundational] • Target: Finding the range algebraically

Find the range of $g(x) = \sqrt{9 - x^2}$ (the upper half of a circle).



7 [Foundational] • Target: Domain and range in applied contexts

A rectangular garden has perimeter 40 m. If x is the width, the length is $20 - x$, and the area is $A(x) = x(20 - x)$.

a State the domain of A in this context (both dimensions must be positive).

b Find $A(6)$ and $A(15)$.

c State the range of A over this domain, to the nearest whole number.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and range in applied contexts

An open-top box is formed from a 12 cm square by cutting squares of side x from each corner, giving volume $V(x) = x(12 - 2x)^2$.

a State the domain of V in this physical context.

b Find $V(1)$ and $V(4)$.

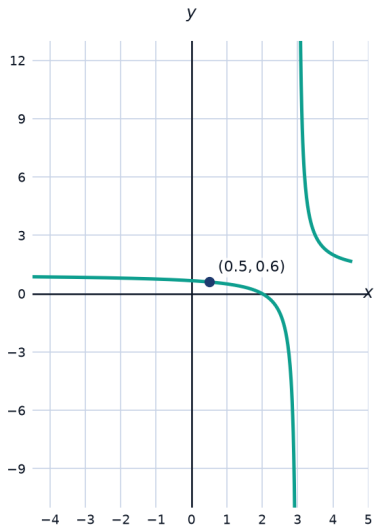
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9 [Intermediate] • Target: Domain of composite and combined functions

Find the domain of $f(x) = \frac{1}{\sqrt{x^2 - 4}}$, writing your answer in interval notation.

10 [Intermediate] • Target: Domain from rational functions with factoring

Find the domain of $f(x) = \frac{x^2 - 4}{x^2 - x - 6}$, factoring the denominator first.



11 [Intermediate] • Target: Domain from rational functions with factoring

Find the domain of $g(x) = \frac{\sqrt{x + 4}}{x - 1}$, combining a root restriction with a denominator restriction.

12 [Intermediate] • Target: Range using transformations

The range of $f(x) = x^2$ is $y \geq 0$. State the range of $g(x) = f(x) - 5 = x^2 - 5$ and of $h(x) = -f(x) = -x^2$, using transformation reasoning.

13 [Intermediate] • Target: Range using transformations

The range of $f(x) = \sqrt{x}$ is $y \geq 0$. State the range of $g(x) = 3\sqrt{x} + 2$.

14 [Intermediate] • Target: Domain and range of piecewise functions

A function is defined by $f(x) = x^2$ for $x < 0$ and $f(x) = 2x + 1$ for $x \geq 0$. State the domain and range.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

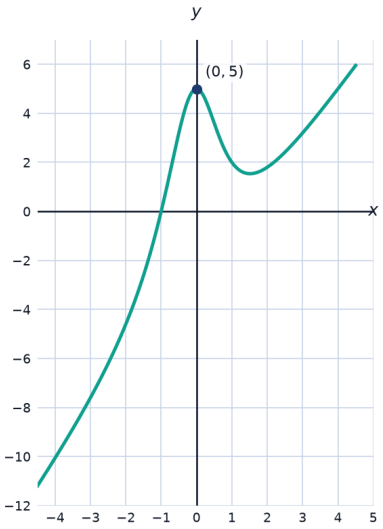
15 [Challenging] • Target: Range using transformations • Cross-Lesson: 5.03 Parametric Equations

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

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16 [Challenging] • Target: Domain and range of piecewise functions • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.



17 [Challenging] • Target: Domain from equations • Cross-Lesson: 6.02 Polar Equations and Graphs

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

18 [Challenging] • Target: Domain and range from graphs • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

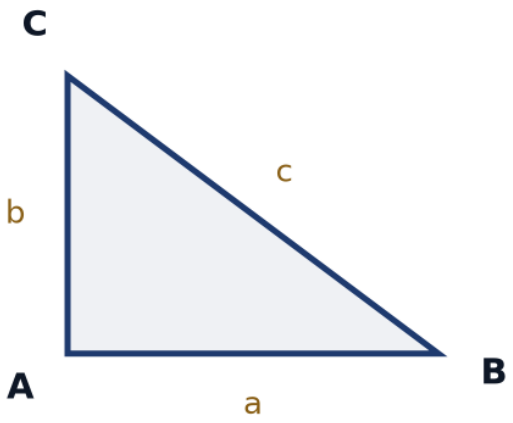
19 [Challenging] • Target: Finding the range algebraically • Cross-Lesson: 7.02 Logarithms and Their Properties

Level 3 · AdvancedLet $R(x) = \frac{2x^2 - 8}{x^2 - 5x + 6}$. (a) Find the domain of $R(x)$. (b) Identify any holes and state their coordinates. (c) Write the equations of all vertical and horizontal asymptotes.

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20 [Challenging] • Target: Domain and range in applied contexts • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Domain from equations

Using the Remainder Theorem, find the remainder when $f(x) = 2x^2 - 1x + 6$ is divided by $x + 3$.

a. -27

b. 28

c. 27

d. 26

22 [Foundational] • Target: Domain and range from graphs

For $f(x) = 2x^3 + 2x^2 - x + 6$, which values are possible rational roots (by the Rational Root Theorem)?

a. $\pm 1, \pm 2$

b. ± 6

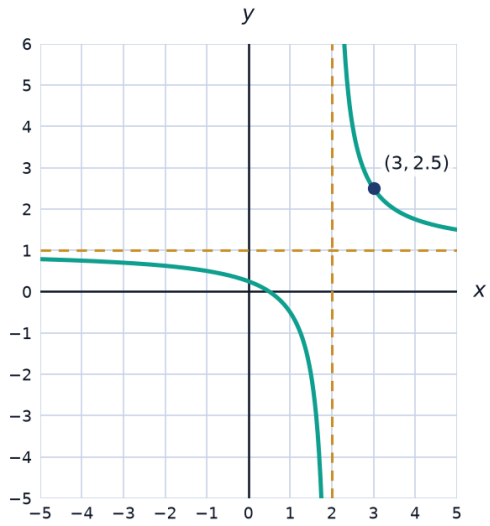
c. ± 12

d. $\pm \frac{1}{2}, \pm 1, \pm \frac{3}{2}, \pm 2, \pm 3, \pm 6$

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23 [Foundational] • Target: Finding the range algebraically

Find all vertical asymptotes of $f(x) = \frac{x^2 + x - 6}{x^3 - 6x^2 + 3x + 10}$.



a. x=2,-1,5

b. x=2

c. x=-1

d. x=-1,5

24 [Foundational] • Target: Domain and range in applied contexts

For $f(x) = \frac{(x+2)(x+1)}{(x+1)(x+3)}$, identify the hole and vertical asymptote.

a. Hole at $x = -3$, vertical asymptote at $x = -1$

b. Hole at $x = -1$, vertical asymptote at $x = -3$

c. Vertical asymptotes at $x = -1$ and $x = -3$, no hole

d. Hole at $x = -2$, vertical asymptote at $x = -3$

25 [Foundational] • Target: Domain of composite and combined functions

For $f(x) = (x - 4)^3(x + 10)$, describe the behavior of the graph at $x = 4$.

a. crosses the x-axis normally (linear)

b. crosses the x-axis, flattening (S-shaped) near the zero

c. touches the x-axis and turns around (does not cross)

d. the graph has a vertical asymptote there

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26 [Foundational] • Target: Domain from rational functions with factoring

As $x \to +\infty$, what happens to $f(x) = 2x^5$?

a. $+\infty$

b. $-\infty$

c. 0

d. Oscillates

27 [Foundational] • Target: Range using transformations

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x + 1}{x - 1}$.

a. $y = 6x$

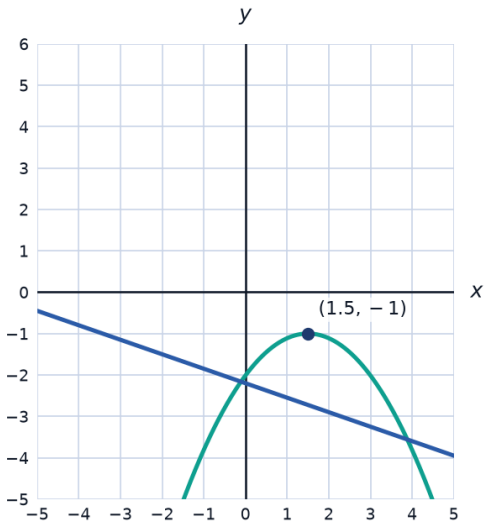
b. $y = x + 6$

c. $y = x + 7$

d. No slant asymptote

28 [Foundational] • Target: Domain and range of piecewise functions

Find the x-coordinate of the hole in the graph of $f(x) = \frac{x^2 + x - 6}{x^2 - 7x + 10}$.



a. x=5

b. x=2

c. No hole exists

d. x=-3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Domain from equations

Find the horizontal asymptote of $f(x) = \frac{x^3+1}{x^2-4}$, or state that none exists.

a. The line $y = x$

b. y=0

c. y=1

d. No horizontal asymptote

30 [Intermediate] • Target: Domain and range from graphs

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$

d. $x = \log_{100}(7)$

31 [Intermediate] • Target: Finding the range algebraically

What is the horizontal asymptote of $f(x) = \frac{1}{x}$?

a. $y = 1$

b. $y = 0$

c. None

d. $x = 0$

32 [Intermediate] • Target: Domain and range in applied contexts

What is the vertical asymptote of $f(x) = \frac{1}{x-0}$?

a. $x = 0$

b. $y = 0$

c. $x = 1$

d. None

33 [Intermediate] • Target: Domain of composite and combined functions

Find the horizontal asymptote of $f(x) = \frac{3}{x^2-4}$.

a. y=0

b. y=-1

c. y=1

d. y=3

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34 [Intermediate] • Target: Domain from rational functions with factoring

Simplify $f(x) = \frac{x^2 + x - 6}{x^2 - 7x + 10}$ and evaluate at $x = 1$.

a. 1

b. 0

c. 2

d. -1

35 [Intermediate] • Target: Range using transformations

What is the degree and leading coefficient of $f(x) = 2x^4 + x^2 - 2$?

a. Degree 5, leading coefficient 2

b. Degree 4, leading coefficient 2

c. Degree 4, leading coefficient -2

d. Degree 3, leading coefficient 2

36 [Intermediate] • Target: Domain and range of piecewise functions

Find the vertical asymptote of $f(x) = \frac{2}{x - 3}$.

a. y=3

b. x=0

c. x=3

d. x=-3

37 [Intermediate] • Target: Domain from equations

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Domain and range from graphs • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$

c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

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39 [Challenging] • Target: Finding the range algebraically • Cross-Lesson: 7.02 Logarithms and Their Properties

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 10)

a. $6 - 24$

b. 22

c. $2 - 64$

d. $6 + 24$

40 [Challenging] • Target: Domain and range in applied contexts • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula.

a. $2 - 64$

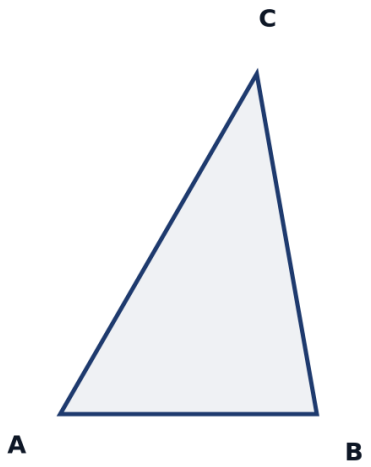
b. $\frac{1}{2}$

c. 22

d. $6 - 24$

41 [Challenging] • Target: Domain of composite and combined functions • Cross-Lesson: 7.04 Applications: Growth and Decay

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.



a. 5.7

b. 6.7

c. 7.7

d. 4.7

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42 [Challenging] • Target: Domain from rational functions with factoring • Cross-Lesson: 8.01 Rational Functions and Asymptotes

In triangle ABC, point D lies on side BC such that AD is an angle bisector of angle A. Let $AB = c$, $AC = b$, $BD = m$, and $CD = n$. Prove that $\frac{m}{n} = \frac{c}{b}$ using a combination of the Sine Rule and the properties of angle bisectors.

a. This statement is a direct consequence of the Sine Rule applied to triangles ABD and ACD.

b. This statement can be proven by applying the Law of Cosines to triangles ABD and ACD and equating expressions for AD^2.

c. This statement is true only if triangle ABC is isosceles with $b = c$.

d. This statement requires the use of vector methods for its proof.

43 [Challenging] • Target: Range using transformations • Cross-Lesson: 8.02 Graphing Rational Functions

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.

a. Consistent and independent (one solution)

b. Consistent and dependent (infinitely many solutions)

c. Inconsistent (no solution)

d. Cannot be determined

44 [Challenging] • Target: Domain and range of piecewise functions • Cross-Lesson: 8.03 Introduction to Limits

Find the projection of $\mathbf{u} = \langle 3, 2 \rangle$ onto $\mathbf{v} = \langle 2, 2 \rangle$.

a. $\left\langle \frac{3}{2}, \frac{5}{2} \right\rangle$

b. $\left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$

c. $\left\langle \frac{7}{2}, \frac{5}{2} \right\rangle$

d. $\frac{5}{4}$

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45 [Challenging] • Target: Domain from equations • Cross-Lesson: 9.01 Arithmetic Sequences and Series

If $\sin\theta = -\frac{20}{29}$ and θ is in Quadrant 3, find $\cos\theta$.

a. $\frac{29}{21}$

b. $\frac{21}{29}$

c. $-\frac{21}{29}$

d. $\frac{20}{29}$

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